

**Subject:** Manual, Two step FVE method

**From** : Jan Mooiman

**Date** : 2026-08-04 21:05:58

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## 1 Description of 1D input file

Five observation points are hard coded in the software, these are located at:

- 1 the centre of the grid.
- 2 at the west boundary.
- 3 at the east boundary.
- 4 halfway to the west boundary.
- 5 halfway to the east boundary.

Extra observation points can be specified in the input file according the description in [section 5](#)

## 2 File description Burgers & Advection-diffusion

An example input file is given in [section 6](#).

**Table 1:** Standard input-file for Burgers and advection-diffusion equation with default settings.

| Keyword     | Default     | Description                                                                                     |
|-------------|-------------|-------------------------------------------------------------------------------------------------|
| logging     | "None"      | "iterations"   "matrix"   "pattern"                                                             |
| [Boundary]  |             |                                                                                                 |
| treg        | 150.0       | Regularization time boundary signal                                                             |
| eps_bc_corr | 0.01        | Boundary correction term to force the boundary value to the given boundary value                |
| bc_signals  | [ · , · ]   | ["constant"   "sine"]                                                                           |
| bc_type     | [ · , · ]   | ["dirichlet"   "borsboom"]                                                                      |
| bc_vals     | [ · , · ]   | [Real value]                                                                                    |
| [Domain]    |             |                                                                                                 |
| Lx          | -           | Length of the domain                                                                            |
| x_origin    | -           | $x$ location of the origin                                                                      |
| [Initial]   |             |                                                                                                 |
| ini_vals    | [ · [, · ]] | [ real value, real value]<br>If one value is given it is assumed constant over the whole domain |
| ini_vars    | "constant"  | ["constant", "linear"]                                                                          |

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| Keyword                       | Default    | Description                                                                              |
|-------------------------------|------------|------------------------------------------------------------------------------------------|
| <b>[Numerics]</b>             |            |                                                                                          |
| dt                            | 5.0        | Time step size [s], if dt == 0: then stationary simulation                               |
| theta                         | 0.5        | Implicitness factor ( $0.5 \leq \theta \leq 1.0$ )                                       |
| c_psi                         | 4.0        | Smoothness factor used for regularization                                                |
| iter_max                      | 25         | Maximum number of nonlinear iterations                                                   |
| eps_newton                    | 1e-12      | Accuracy of the Newton solver (non-linear)                                               |
| eps_bicgstab                  | 1e-12      | Accuracy of the linear solver                                                            |
| eps_abs_function              | 0.01       | Smoothing factor for the absolute-function                                               |
| linear_solver                 | "bicgstab" | Type of linear solver "bicgstab", <b>"multigrid"</b>                                     |
| regularization_init           | false      | true, false                                                                              |
| regularization_iter           | false      | true, false                                                                              |
| regularization_time           | false      | true, false                                                                              |
| <b>[Output]</b>               |            |                                                                                          |
| dt_his                        | dt         | Time interval to write the history file, [s]                                             |
| dt_map                        | 60.0       | Time interval to write the map file, [s]                                                 |
| dt_screen                     | 60.0       | Time interval to write the time step to the screen, [s]                                  |
| <b>[Physics]</b>              |            |                                                                                          |
| do_convection                 | false      | true, false                                                                              |
| do_viscosity                  | false      | true, false                                                                              |
| viscosity                     | 0.01       | Diffusion/Viscosity coefficient                                                          |
| <b>[Time]</b>                 |            |                                                                                          |
| tunit                         | "s"        | Unit of the start and stop time. Values seconds "s", minutes "m", hours "h" and days "d" |
| start                         | 0.0        | Start time of the simulation, [tunit]                                                    |
| stop                          | 60.0       | Stop time of the simulation, [tunit]                                                     |
| <b>[ [ObservationPoint] ]</b> |            |                                                                                          |
| x                             | -          | $x$ -coordinate of the observation point                                                 |
| y                             | -          | $y$ -coordinate of the observation point                                                 |
| name                          | "..."      | Name of observation point between double quotes                                          |

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| Keyword | Default | Description |
|---------|---------|-------------|
|---------|---------|-------------|

### 3 File description Wave 1D

An example input file is given in [section 7](#).

**Table 2:** Standard input-file for wave-1d equation with default settings.

| Keyword           | Default   | Description                                                                                                                                                                                                                                                                                |
|-------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| logging           | "None"    | "iterations"   "matrix"   "pattern"                                                                                                                                                                                                                                                        |
| <b>[Boundary]</b> |           |                                                                                                                                                                                                                                                                                            |
| treg              | 150.0     | Regularization time boundary signal                                                                                                                                                                                                                                                        |
| eps_bc_corr       | 0.01      | Boundary correction term to force the boundary value to the given boundary value                                                                                                                                                                                                           |
| bc_type           | [ · , · ] | ["borsboom"   "mooiman"]                                                                                                                                                                                                                                                                   |
| bc_vals           | [ · , · ] | [Real values]                                                                                                                                                                                                                                                                              |
| bc_vars           | [ · , · ] | ["zeta"   "q"]                                                                                                                                                                                                                                                                             |
| <b>[Domain]</b>   |           |                                                                                                                                                                                                                                                                                            |
| Lx                | -         | Length of the domain, [m]                                                                                                                                                                                                                                                                  |
| depth             | 10.0      | Bathymetry below reference level [m]                                                                                                                                                                                                                                                       |
| geomtry_type      | [ · ]     | "UniformGeometry"<br>Uniform depth profile<br>"WavyBedGeometry"<br>Wavy bed geometry<br>"SlopedBedGeometry"<br>Sloped bed geometry (Chézy flow)<br>"SlopedWavyBedGeometry"<br>Sloped wavy bed geometry<br>"WeirBorsboom2000Geometry"<br>Profile of a weir<br>"Shoal"<br>Profile of a shoal |
| <b>[Initial]</b>  |           |                                                                                                                                                                                                                                                                                            |
| ini_vars          | "-"       | ["zeta_gauss_hump"]                                                                                                                                                                                                                                                                        |
| gauss_app         | "-"       | Twice the amplitude of the Gaussian hump at the boundary                                                                                                                                                                                                                                   |
| gauss_mu          | "-"       | Location of the top of the Gaussian hump                                                                                                                                                                                                                                                   |
| gauss_sigma       | "-"       | Width of the Gaussian hump                                                                                                                                                                                                                                                                 |

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| Keyword                       | Default    | Description                                                                              |
|-------------------------------|------------|------------------------------------------------------------------------------------------|
| <b>[Numerics]</b>             |            |                                                                                          |
| dt                            | -          | Time step size [s], if dt == 0: then stationary simulation                               |
| dx                            | -          | Grid size, [m]                                                                           |
| theta                         | 0.5        | Implicitness factor ( $0.5 \leq \theta \leq 1.0$ )                                       |
| c_psi                         | 4.0        | Smoothness factor used for regularization                                                |
| iter_max                      | 25         | Maximum number of nonlinear iterations                                                   |
| eps_newton                    | 1e-12      | Accuracy of the Newton solver (non-linear)                                               |
| eps_bicgstab                  | 1e-12      | Accuracy of the linear solver                                                            |
| eps_abs_function              | 0.01       | Smoothing factor for the absolute-function                                               |
| linear_solver                 | "bicgstab" | Type of linear solver "bicgstab", "multigrid"                                            |
| regularization_init           | false      | true, false                                                                              |
| regularization_iter           | false      | true, false                                                                              |
| regularization_time           | false      | true, false                                                                              |
| <b>[Output]</b>               |            |                                                                                          |
| dt_his                        | dt         | Time interval to write the history file, [s]                                             |
| dt_map                        | 60.0       | Time interval to write the map file, [s]                                                 |
| dt_screen                     | 60.0       | Time interval to write the time step to the screen, [s]                                  |
| <b>[Physics]</b>              |            |                                                                                          |
| do_convection                 | false      | true, false                                                                              |
| do_viscosity                  | false      | true, false                                                                              |
| viscosity                     | 0.01       | Diffusion/Viscosity coefficient                                                          |
| do_source                     | false      | true, false                                                                              |
| src_type                      | "none"     | "none", "quadratic"                                                                      |
| <b>[Time]</b>                 |            |                                                                                          |
| tunit                         | "s"        | Unit of the start and stop time. Values seconds "s", minutes "m", hours "h" and days "d" |
| start                         | 0.0        | Start time of the simulation, [tunit]                                                    |
| stop                          | 60.0       | Stop time of the simulation, [tunit]                                                     |
| <b>[ [ObservationPoint] ]</b> |            |                                                                                          |
| x                             | -          | $x$ -coordinate of the observation point                                                 |
| y                             | -          | $y$ -coordinate of the observation point                                                 |
| name                          | "..."      | Name of observation point between double quotes                                          |

## 4 Description of 2D input file

Nine observation points are hard coded in the software, these are located at:

- 1 the centre of the grid.
- 2 in the middle of the north boundary.
- 3 in the middle of the east boundary.
- 4 in the middle of the south boundary.
- 5 in the middle of the west boundary.
- 6 in the North-East corner of the grid.
- 7 in the South-East corner of the grid.
- 8 in the South-West corner of the grid.
- 9 in the North-West corner of the grid.

Extra observation points can be specified in the input file according the description in [section 5](#).

## 5 File description

Items in the color red are only to be used in the development stage of the software. An example input file is given in [section 8](#).

**Table 3:** Standard input-file with default settings.

| Keyword           | Default           | Description                                                                      |
|-------------------|-------------------|----------------------------------------------------------------------------------|
| logging           | "None"            | "iterations"   "matrix"   "pattern"                                              |
| <b>[Boundary]</b> |                   |                                                                                  |
| treg              | 150.0             | Regularization time boundary signal                                              |
| eps_bc_corr       | 0.01              | Boundary correction term to force the boundary value to the given boundary value |
| bc_type           | [ · , · , · , · ] | ["free-slip"   "no-slip"   "borsboom"   "mooiman" ]                              |
| bc_vars           | [ · , · , · , · ] | ["-"   "zeta"   "q"]                                                             |
| bc_vals           | [ · , · , · , · ] | [Real value]                                                                     |
| <b>[Domain]</b>   |                   |                                                                                  |
| bed-level-file    | -                 | Name of bed-level-file between double quotes                                     |
| mesh-file         | -                 | Name of mesh file point between double quotes                                    |

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| Keyword             | Default    | Description                                                                               |
|---------------------|------------|-------------------------------------------------------------------------------------------|
| <b>[Initial]</b>    |            |                                                                                           |
| ini_vars            | None       | ["zeta", "zeta_gauss_hump"   "zeta_constant"   "zeta_gauss_hump_x"   "zeta_gauss_hump_y"] |
| gauss_amp           | 0.0        |                                                                                           |
| gauss_mu            | 0.0        |                                                                                           |
| gauss_mu_x          | 0.0        |                                                                                           |
| gauss_mu_y          | 0.0        |                                                                                           |
| gauss_sigma         | 350.0      |                                                                                           |
| gauss_sigma_x       | 350.0      | Optional if "gauss_sigma" is not specified                                                |
| gauss_sigma_y       | 350.0      | Optional if "gauss_sigma" is not specified                                                |
| <b>[Numerics]</b>   |            |                                                                                           |
| dt                  | 5.0        | Time step size [s], if dt == 0: then stationary simulation                                |
| theta               | 0.501      | Implicitness factor (0.5 <= theta <= 1.0)                                                 |
| c_psi               | 4.0        | Smoothness factor used for regularization                                                 |
| iter_max            | 25         | Maximum number of nonlinear iterations                                                    |
| eps_newton          | 1e-12      | Accuracy of the Newton solver (non-linear)                                                |
| eps_bicgstab        | 1e-06      | Accuracy of the linear solver                                                             |
| eps_abs_function    | 0.01       | Smoothing factor for the absolute-function                                                |
| linear_solver       | "bicgstab" | Type of linear solver "bicgstab", "multigrid"                                             |
| regularization_init | false      | true, false                                                                               |
| regularization_iter | false      | true, false                                                                               |
| regularization_time | false      | true, false                                                                               |
| <b>[Physics]</b>    |            |                                                                                           |
| do_linear_waves     | true       | true, false                                                                               |
| do_continuity       | true       | true, false                                                                               |
| do_q_equation       | true       | true, false                                                                               |
| do_r_equation       | true       | true, false                                                                               |
| do_convection       | false      | true, false                                                                               |
| do_bed_shear_stress | false      | true, false                                                                               |
| do_viscosity        | false      | true, false                                                                               |
| chezy_coefficient   | 25.0       | Bed friction coefficient                                                                  |
| viscosity           | 0.01       | Diffusivity coefficient                                                                   |
| <b>[Output]</b>     |            |                                                                                           |
| dt_his              | dt         | Time interval to write the history file, [s]                                              |
| dt_map              | 60.0       | Time interval to write the history file, [s]                                              |
| dt_screen           | 60.0       | Time interval to write the time step to the screen, [s]                                   |
| <b>[Time]</b>       |            |                                                                                           |
| tunit               | "s"        | Unit of the start and stop time. Values seconds "s", minutes "m", hours "h" and days "d"  |
| start               | 0.0        | Start time of the simulation, [s]                                                         |
| stop                | 60.0       | Stop time of the simulation, [s]                                                          |

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| Keyword                       | Default | Description                                     |
|-------------------------------|---------|-------------------------------------------------|
| [[ <b>ObservationPoint</b> ]] |         |                                                 |
| x                             | -       | $x$ -coordinate of the observation point        |
| y                             | -       | $y$ -coordinate of the observation point        |
| name                          | "..."   | Name of observation point between double quotes |

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## 6 Example input 1D file Advection-diffusion

```

Logging = "None" # "iterations", "matrix", "pattern"

[Boundary] # west, east
treg      = 1800.0 # Regularization time boundary signal [s]
eps_bc_corr = 0.01
bc_signals = ["constant", "constant"]
bc_type    = ["dirichlet", "borsboom"]
bc_vals    = [2.0, 0.0]

[Domain]
Lx      = 20000.0 # Domain length [m]
x_origin = 0.0 # Origin of the x-coordinate [m]

[Initial]
ini_vals = [1.0]
ini_vars = ["constant"]

[Numerics]
dt      = 1.0 # Time step size [s], if dt == 0: then stationary problem
dx      = 1.0 # Grid size [m]
c_psi   = 4.0
theta   = 0.50 # Implicitness factor (0.5 <= theta <= 1.0)
iter_max = 25.0 # Maximum number of non-linear iterations
eps_newton = 1e-12
eps_bicgstab = 1e-12
eps_abs_function = 0.01
regularization_init = true
regularization_iter = false
regularization_time = true

[Output]
dt_hist = 1.0
dt_map  = 300.0
dt_screen = 300.0

[Physics]
do_convection = true
do_viscosity  = true
viscosity     = 0.1
do_source     = false
src_type      = "none"

[Time]
tunit = "s" # "s", "m", "h", "d"
tstart = 0.0
tstop  = 14400.0

# Hard coded observation points are not listed

[[ObservationPoint]]
x      = 2500
y      = 0
name   = "_a"

[[ObservationPoint]]

```



```
x      = 7500
y      = 0
name = "_b"

[[ObservationPoint]]
x      = 12500
y      = 0
name = "_c"

[[ObservationPoint]]
x      = 17500
y      = 0
name = "_d"
```

## 7 Example input 1D file Wave 1D

```
[Boundary]
bc\_type = ["borsboom", "borsboom"] # Type "mooiman", "borsboom"
bc\_vars = ["q", "zeta"]             # Incoming signal "q" or "zeta"
bc\_vals = [0.0, 0.0]                # The inflow discharge
treg = 600.0                         # Regularization time on boundary
eps\_bc\_corr = 1.0e-2

[Domain]
Lx = 36000.0 # Domain length [m]
depth = 10.0 # bathymetry below reference level [m]
geometry\_type = "UniformGeometry"

[Initial]
ini\_vars = ["zeta\_gauss\_hump"]
gauss\_mu = 9000.0 # location of the top of the gaussian hump
gauss\_sigma = 700.0 # location of the top of the gaussian hump
gauss\_amp = 0.02 # twice the amplitude of the gaussian hump at the boundary

[Numerics]
dt = 10.0 # Time step size [s], if dt == 0: then stationary problem
dx = 10.0 # Grid size [m]
iter\_max = 25 # Maximum number of nonlinear iterations
theta = 0.501 # Implicitness factor (0.5 <= theta <= 1.0)
eps\_newton = 1.0e-12
eps\_bicgstab = 1.0e-12

[Output]
dt\_his = 10.0 # [s]
dt\_map = 1200.0 # [s]
dt\_screen = 300.0 # [s]

[Physics]
g = 10.0 # Gravitational acceleration
do\_convection = false # Include momentum convection term

[Time]
tstart = 0.0 # Start time [s]
tstop = 43200.0 # End time [s]
```

## 8 Example input 2D file

```

logging = "None" # "iterations", "matrix", "pattern"

[Boundary] # north, east, south, west
treg      = 150.0 # Regularization time boundary signal
eps_bc_corr = 0.01
bc_type    = ["borsboom", "borsboom", "borsboom", "borsboom"] # Type "neumann", "dirichlet", "borsboom"
bc_vars    = ["zeta", "zeta", "zeta", "zeta"]
bc_absorbing = [true, true, true, true]
bc_vals    = [0.0, 0.0, 0.0, 0.0]

[Domain]
mesh_file    = "geometry/62x62_6x6km_net.nc"
bed_level_file = "geometry/62x62_6x6km_flat.dep"

[Initial]
ini_vars     = ["zeta", "zeta_gauss_hump"]
Gauss_amp    = 1.0
Gauss_mu     = 0.0
Gauss_mu_x   = 0.0
Gauss_mu_y   = 0.0
Gauss_sigma  = 350.0
Gauss_sigma_x = 350.0
Gauss_sigma_y = 350.0

[Numerics]
dt           = 5.0 # Time step size [s], if dt == 0: then stationary problem
theta       = 0.501 # Implicitness factor (0.5 <= theta <= 1.0)
c_psi       = 4.0
iter_max    = 25 # Maximum number of nonlinear iterations
eps_newton  = 1e-12
eps_bicgstab = 1e-06
eps_abs_function = 0.01
linear_solver = "bicgstab" # "bicgstab", "multigrid"
regularization_init = false
regularization_iter = false
regularization_time = false

[Physics]
do_linear_waves = true
do_continuity   = true
do_q_equation   = true
do_r_equation   = true
do_convection   = false
do_bed_shear_stress = false
do_viscosity    = false
chezy_coefficient = 25.0
viscosity       = 100.0

[Output]
dt_his = 0.01
dt_map = 30.0

[Time]
tunit = "s"
tstart = 0.0
tstop = 1800.0

```

